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The Long-Term Effect of Biochar Amendment on Soil Biochemistry and Phosphorus Availability of Calcareous Soils

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Abstract: Biochar soil amendment has proven to be beneficial for soil fertility regulation, while the long-term mechanisms under different soil conditions remain largely unexplored. This study was based on two long-term field experiments in calcareous soils to explore the effect of biochar on fertility and phosphorus availability, as well as potential mechanisms. Experiments were conducted on Fluvo-aquic soil (6 years) and Cinnamon soil (5 years), respectively. Biochar was applied once at 20 t/ha level and mixed with the topsoil. We evaluated soil fertility and phosphorus fractions with and without biochar amendment and analyzed the related extracellular enzyme activity and functional gene abundance. The results indicate a persistent increase in crop yield by 10.8–24.3% in calcareous soils after 5–6 years of treatment. Biochar enhanced soil fertility, including increased SOC, available phosphorus, and potassium content. Moreover, it boosted the activity of extracellular enzymes related to soil carbon, nitrogen, and phosphorus cycling. This effect may be associated with the stability of morphological features and mineral element regulation on biochar surfaces. Biochar soil amendment led to a long-term increase in phosphorus availability, of which available phosphorus increased by 13.1–49.1%. Potential mechanisms included an increase in labile phosphorus forms and biological mechanisms of raising alkaline phosphatase activities and increasing the copy numbers of the phosphatase genes (*phoD*, *phoX*, and *nifH*). In conclusion, biochar soil amendment resulted in sustained improvement in soil fertility in calcareous soils and consistently increased phosphorus availability via biochemical mechanisms.

Keywords: crop yield; extracellular enzymes activity; *phoD* gene; phosphorus fraction; ^{31}P-MNR

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1. Introduction

Biochar soil amendment holds the benefit of improving soil fertility and enhancing crop productivity. Many studies reported that biochar promotes the formation of soil aggregate, augments nutrient and water content [1,2], and increases crop yield [3]. Biochar remains stable in soil, so recent research has centered on the long-term effect of biochar treatment on crop production. Studies using ^{13}C and ^{14}C isotopes have shown that they can persist in soils for centuries [4]. Biochar soil amendment continually enhances soil fertility through abiotic and biotic mechanisms. Regarding chemical mechanisms, the

natural aging of biochar results in changes to surface morphologies, such as fragmentation caused by temperature variations or organisms' activities, which can lead to an increase in the contact interface with the soil, which will promote the release of materials in biochar [5]. Aging introduces oxygen-containing functional groups onto biochar surfaces, thereby increasing the negative surface charge and enhancing the soil CEC [6]. In terms of physical mechanisms, a meta-analysis demonstrated that biochar soil amendment supports soil aggregate formation in long-term field experiments, improving soil aeration and water retention capacity [7]. For biotic mechanisms, biochar amendment can have a lasting effect on soil microbial abundance and communities [8]. It also significantly enhances the soil microbial biomass carbon content and extracellular enzyme activities such as dehydrogenase and alkaline phosphatase in long-term studies [9]. The persistent impact of biochar on soil fertility contributes to long-term benefits for crop productivity [10,11].

Biochar amendment holds the potential for enhancing phosphorus (P) use efficiency. Biochar amendment is a method of recycling nutrients in agricultural systems, notably P. Meta-analysis studies have shown that biochar increases both soil P availability and its uptake by crops [3,12]. In the short term, biochar can directly supply nutrients to soil upon application. However, the long-term effect involves multiple mechanisms. This was evidenced in a decade-long field study [13], in which a woody biochar with a low initial P content was added to a Ferralsol, resulting in a persistent increase in soil P availability. Therefore, biochar amendment could modulate the transformation of soil P fractions via a network of biological and physicochemical mechanisms [14,15]. Phosphates readily bind with cations such as Ca^{2+}, Mg^{2+}, Fe^{3+}, and Al^{3+}, with calcium phosphates serving as the primary stabilizing mechanism in calcareous soils [16]. Biochar amendment can reduce soil P fixation by influencing soil P fractions and soil reactions. The functional groups and electrostatic adsorption sites on the biochar surface can interact with organic components and soil minerals to form organo-mineral phases [17]. This reaction is one of the potential mechanisms for soil structure formation and nutrient cycling under both short and long-term conditions. The long-term impact on P availability is largely due to the mechanism of microbial activation. Recently, phosphate-solubilizing microbes such as Streptomyces, Agrobacterium, Rhizopus, Pseudomonas, and Aspergillus have been studied for their roles in soil P activation [18]. Vanek and Lehmann [19] found that biochar can interact with arbuscular mycorrhizae, facilitating soil P dissolution. Zhou et al. [20] reported that leaf biochar increased soil P availability by boosting the diversity and abundance of P-solubilizing bacteria, such as *Singulisphaera* and *Planctomyces*. These associated soil microbes assist in the conversion of soil P into bioavailable forms in the soil solution, such as orthophosphate ($H_2PO_4^-$ or HPO_4^{2-}). Soil organic P mineralization is directly facilitated by extracellular enzymes, including acidic phosphatases, alkaline phosphatases, nucleotidases, and phytases [21], which contribute to the release of orthophosphate from organic P forms. Alkaline phosphatase, a non-specific enzyme, has been reported to contribute to P bioavailability in soil. Yang et al. [22] have found that biochar amendment significantly increases alkaline phosphatase activity. Biochar soil amendment also shows the potential to increase the copy numbers of P-functional genes [23,24], such as *phoA*, *phoD*, and *phoX*, which could enhance soil P bioavailability.

The role of biochar in soil nutrient regulation and crop production varies between soils due to factors such as soil acidity and fertility [25,26]. The long-term response of soil-crop systems under varying soil conditions, especially, and the potential biochemical mechanisms remain largely unknown [27]. Therefore, this study conducted two long-term field experiments in two specified calcareous soils (Fluvo-aquic soil and Cinnamon). We aimed to explore the long-term effect of biochar on soil fertility and P availability in calcareous soils, as well as potential mechanisms. Our hypothesis is that biochar is

stable in the soil, and its porous structure and large specific surface will consistently preserve soil nutrients. Biochar will also regulate soil nutrients, including P availability, via a series of biochemical mechanisms. This study can accurately reflect the actual effect of biochar on agricultural production, provide a theoretical basis for the efficient use of nutrients in agricultural soils, and have a certain value from the perspective of sustainable agricultural development.

2. Materials and Methods

2.1. Experimental Sites

This study was based on two field experiments. The first experiment, Site I, was conducted in 2013 and located in Yaozhuang village (114.064615° E, 35.256756° N), Xinxiang City, Henan Province, China. The mean annual temperature and precipitation are 14.0 °C and 573 mm. The local cropping system rotates between wheat and maize; maize is planted in June, and wheat is grown in October. The soil is Fluvo-aquic soil with basic properties shown in Table 1.

Table 1. The basic properties of Fluvo-aquic and Cinnamon soils.

Soil	pH	SOC g kg^{-1}	Total N g kg^{-1}	Available P mg kg^{-1}	Available K mg kg^{-1}
Fluvo-aquic soil	7.98	7.45	0.81	68.5	120.5
Cinnamon	8.05	3.01	0.49	13.6	99.8

The second experiment, Site II, was initiated in 2014 in Hongquan village (112.659242° E, 35.050319° N), Jiyuan City, Henan Province, China. The mean annual temperature and precipitation are 14.5 °C and 568 mm. The local cropping system also rotates between wheat and maize. The local soil is Cinnamon, with basic properties shown in Table 1.

2.2. Experiment Design

Wheat straw biochar (WBC) and maize straw biochar (MBC) were used in this study. The two biochars were produced at a pyrolysis temperature of approximately 450 °C. Prior to application, the biochars were ground to pass through a 2 mm sieve. The biochar properties are shown in Table 2.

Table 2. The properties of WBC and MBC.

Biochar	C g kg^{-1}	N g kg^{-1}	P g kg^{-1}	K mg kg^{-1}	pH	Surface Area m^2 g^{-1}
WBC	467	6.0	11.5	26.1	9.43	15.3
MBC	496	10.2	9.05	33.4	9.03	26.4

Field Experiment I commenced in June 2013 and ran until 2019. This experiment included three treatments: CK (no biochar addition), WBC (WBC applied at 20 t ha^{-1}), and MBC (MBC applied at 20 t ha^{-1}). And a randomized block design with 3 replicates.

Field Experiment II started in June 2014 and ran until 2019. This experiment only featured two treatments: CK (no biochar) and WBC (20 t ha^{-1}). Both experiments employed plots of 20 m^2 (width × length of 4 m × 5 m). Biochar was applied once in each experiment, with no additional biochar applied in subsequent years. The biochar was applied to a depth of 15 cm and mixed with the upper soil layer via plowing. For both experiments, chemical fertilizers of N, P_2O_5, and K_2O were applied at 225 kg ha^{-1}, 90 kg ha^{-1}, and 90 kg ha^{-1}, respectively, in both wheat and maize seasons, which was according to the local fertilization practice, and the fertilization rate was consistent every year during the experiment period.

Crop straw was fully returned after harvest, and no additional manure or organic fertilizer was applied. The other field management practices followed local conventional protocols. The crop yield for each treatment was determined through manual harvesting.

2.3. Soil Sampling and Analysis

Soil sampling was performed in September 2019, coinciding with the maize harvest. Topsoil samples (0–15 cm) were collected with five randomly chosen soil cores forming a composite sample in each plot. These samples were sifted to remove plant residues and gravel and then ground to pass through a 2 mm sieve. Then, one portion was air dried at room temperature for nutrient content analysis, one portion was preserved at 4 °C for extracellular enzyme activity analysis, and the rest was stored at −80 °C for quantification of P cycling functional genes.

The physicochemical properties such as soil pH, available N, available P, and available K content were analyzed following Lu [28]; pH was measured in 1:2.5 soil/water suspensions; available N was determined by alkaline diffusion method; available P was determined by Olsen method; available K was determined by flame spectrophotometry. The SOC and total N content were determined using Vario MAX CNS Analyzer (Elementar, 2003). Soil aggregates were determined following Stemmer et al. [29], which were firstly dispersed by low-energy sonication and then separated by wet sieving and centrifuging. The soil aggregates were separated into four particle sizes (>2 mm, 2–0.25 mm, 0.25–0.053 mm, and <0.053 mm), and mean weight diameter (MWD) was calculated.

2.4. Soil P Fractions

Soil inorganic P fractions were analyzed following the continuous extraction fractionation scheme for calcareous soils outlined by Lu [28]. The inorganic P forms of Ca_2-P, Ca_8-P, Fe-P, O-P, and Ca_{10}-P were measured.

The organic fraction of soil P was analyzed using ^{31}P-NMR spectroscopy. The P fraction was extracted according to the protocol outlined by Noack et al. [30]. Briefly, 1.00 g of air-dried soil was extracted with 30 mL of 50 mmol L^{-1} EDTA and 0.25 mol L^{-1} NaOH solution. Methylene disphosphonic acid (MDPA) was added at 2.5 mg L^{-1} as the internal standard, followed by freeze-drying. Subsequently, 300 mg of the extract was re-dissolved in 0.6 mL of deuterium oxide and 0.1 mL of 10 mol L^{-1} NaOH solution. This solution underwent ultrasonic oscillation for 30 min at room temperature, a 5 min equilibration period, and then centrifugation for 15 min and 14,000 rpm. The supernatants were transferred to a 5 mm NMR tube in a double-resonance probe head. Solution ^{31}P-NMR spectra were acquired at 20 °C on a Bruker Avance 400 NMR spectrometer, the parameters set to: resonance frequency of SF, 161.975 MHz; spectral width of SWH, 15,000–25,000 Hz; and recycle delay time of dl, 2.0 s.

Organic P resonances were identified by the chemical shift deviation relative to the dominant orthophosphate peak (5.5–5.6 ppm), and other P-form peaks were assigned based on literature results [31,32]. Resonances of inorganic P forms of polyphosphate and orthophosphate, as well as organic forms of phosphonate, orthophosphate diesters, choline phosphate, IHP (inositol hexaphosphate), and other monoesters were identified. The relative proportions of each P compound were calculated by integration using TopSpin 4.0 software (Brucker Biospin GmbH, Rheinstetten, Germany). Each P fraction was calculated based on the relative ratio of integrated MDPA (16.50 ppm).

2.5. Quantification of Phosphorus Cycling Functional Genes

DNA was extracted using a Fast DNA isolation kit (MP Biomedicals, Santa Ana, CA, USA) from the frozen soil samples. The genes of *phoD*, *phoX*, and *nifH* were amplified using the primers reported by Sakurai et al. [33], Ragot et al. [34], and Zhang et al. [35],

respectively. The amplification conditions were: for *phoD*, initial denaturation at 94 °C for 4 min, followed 40 cycles of 94 °C for 45 s, 57 °C for 30 s, 72 °C for 1 min, and 72 °C for 8 min; for phoX, initial denaturation at 95 °C for 3 min, followed 35 cycles of 95 °C for 30 s, 60 °C for 30 s, 72°C for 30 s, and 72°C for 10 min; and for *nifH*, initial denaturation at 95 °C for 5 min, followed 32 cycles of 95 °C for 45 s, 55 °C for 50 s, 72 °C for 45 s, and 72 °C for 10 min.

The qPCR reactions were performed using the ABI PRISM® 7900HT sequence detection system (Thermo Fisher Scientific, Waltham, MA, USA). The standard curves were generated by tenfold serial dilutions of plasmids containing fragments of the *phoD*, *phoX*, and *nifH* genes. The amplification efficiency exceeded 90%, with an R2 value greater than 0.98.

2.6. Soil Extracellular Enzymes Activity

The activities of 7 extracellular enzymes were determined using the enzyme-linked immunosorbent assay (ELISA) kits (Shanghai Jingkang Bioengineering Co., Ltd. Shanghai, China), including the carbon cycling-related enzymes of β-glucosidase (BG), β-xylosidase (BX) and β-(D)-cellobiosidase (CBH); the nitrogen cycling-related enzymes of N-acetyl-β-glucosaminidase (NAG) and leucine aminopeptidase (LAP); and the phosphorus cycling-related enzymes of acidic phosphatase (ACP) and alkaline phosphatase (ALP).

2.7. SEM-EDS Analysis Biochar

The surface structure and elemental composition analysis of both fresh (selected before the experiment) and aged (selected in 2019 from soils of experiment I) biochar were determined using a scanning electron microscope equipped with energy dispersive spectroscopy (SEM-EDS, Hitachi SU3400, Tokyo, Japan).

2.8. Calculations and Statistics

All data are presented as mean ± standard deviations (n = 3). Statistical analyses were performed using IBM SPSS Statistics 26.0 (SPSS Inc., Chicago, IL, USA). The differences between treatments in Fluvo-aquic soil were analyzed using ANOVA, followed by an LSD test ($p < 0.05$). A t-test was used for Cinnamon soil ($p < 0.05$).

3. Results

3.1. Crop Yield

As presented in Figure 1, biochar had a persistent impact on crop yield in both Fluvo-aquic soil and Cinnamon soil. In Fluvo-aquic soil, the maize yield with WBC treatment was significantly increased by 11.6–24.3%, compared to CK, except in the third year, the MBC treatment only significantly increased maize yield by 10.8% in the sixth year. The WBC treatment also significantly increased wheat yield by 13.7–21.2%, except in the third and fifth years; the MBC treatment significantly increased wheat yield by 16.1% in the fourth year, respectively. In Cinnamon soil, the WBC treatment significantly increased maize yield by 14.2% and 20.4% compared to CK in the third and fifth years, respectively, and increased wheat yield by 11.1% and 13.7% in the second and fifth years, respectively. Consequently, the long-term crop yield enhancement effect of biochar in calcareous soil was confirmed, although it varied with the soil conditions, biochar types, and crop types.

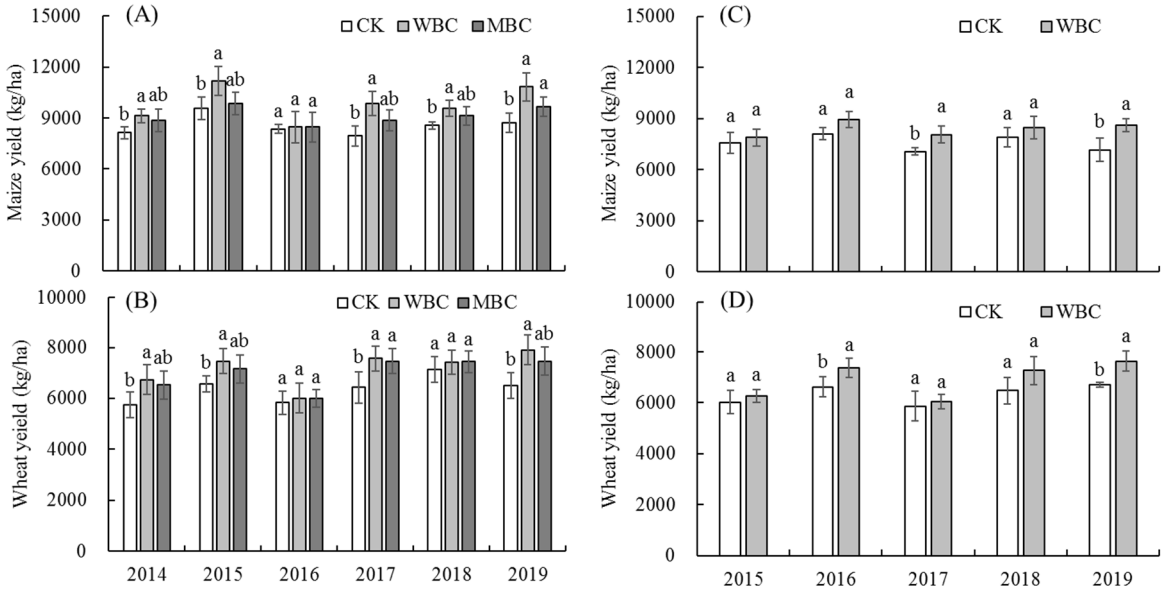


Figure 1. The long-term effect of biochar amendment on maize and wheat yield. Note: (**A**,**B**) represent the maize and wheat yield in Fluvo-aquic soil, and (**C**,**D**) represent the maize and wheat yield in Cinnamon soil, respectively. Different lowercase letters on the column represent the significant difference between treatments ($p < 0.05$).

3.2. Soil Fertility

3.2.1. Soil Physicochemical Properties

Biochar treatments enhanced soil fertility both in the short and long term following a one-time amendment in calcareous soil, particularly increasing SOC, available P, and available K contents (Table 3). In the short term, biochar treatment significantly increased SOC, Available N, Available P, and Available K content, both in Fluvo-aquic soil and Cinnamon soil. In the long term, biochar showed no significant effect on the pH of either calcareous soil. However, it significantly increased aggregate MWD by 53.1–73.5% in Fluvo-aquic soil and 64.7% in Cinnamon soil. The biochar treatment significantly increased SOC content by 16.2–16.3% in Fluvo-aquic soil and 38.6% in Cinnamon soil. The available N content significantly increased in both soils with biochar treatment compared to CK. No significant difference in total P between treatments for either calcareous soils was detected. However, soil available P content was increased by 42.6 mg kg^{-1} and 11.4 mg kg^{-1} with WBC and MBC treatments in Fluvo-aquic soil and by 2.3 mg kg^{-1} with WBC significantly in Cinnamon soil. The soil available K content increased by 110.9–117.6 mg kg^{-1} in Fluvo-aquic soil and 58.9 mg kg^{-1} in Cinnamon soil.

3.2.2. Soil Extracellular Enzyme Activities

In Fluvo-aquic soil, WBC treatment significantly increased BX and CBH activities by 26.8% and 28.5%, respectively, without affecting BG activity, compared to CK (Table 4). However, MBC treatment showed no significant effect. In Cinnamon soil, WBC treatment did not influence CBH and BG activities but increased BX activity by 16.4% compared to CK. Biochar soil amendment significantly increased LAP activity in both Fluvo-aquic soil and Cinnamon soil but had no significant effect on NAG activity. In Fluvo-aquic soil, both MBC and WBC significantly increased ALP activity by 23.9% and 21.6%, respectively. In Cinnamon soil, WBC treatment increased ALP activity by 24.2% compared to CK. Biochar did not significantly affect ACP activity in either of the calcareous soils. Therefore, biochar amendment enhances soil extracellular enzyme activity in calcareous soils, but it also varies with soil conditions.

Table 3. Effect of biochar on the physicochemical properties of soils.

	Soil	Treatment	pH (H_2O)	MWD (mm)	SOC (g kg^{-1})	Total N (g kg^{-1})	Total P (g kg^{-1})	Available N (mg kg^{-1})	Available P (mg kg^{-1})	Available K (mg kg^{-1})
Short-term effect	Fluvo-aquic soil	CK	8.14 ± 0.08 a	0.42 ± 0.07 b	7.38 ± 0.21 b	0.80 ± 0.02 a	1.17 ± 0.15 a	79.62 ± 7.99 b	65.27 ± 6.65 b	108.64 ± 14.62 c
		WBC	8.22 ± 0.12 a	0.58 ± 0.09 a	8.55 ± 0.64 a	0.83 ± 0.04 a	1.33 ± 0.11 a	110.28 ± 12.62 a	117.66 ± 8.97 a	251.66 ± 22.55 a
		MBC	8.15 ± 0.11 a	0.55 ± 0.12 ab	8.53 ± 0.42 a	0.84 ± 0.04 a	1.24 ± 0.17 a	98.67 ± 10.75 a	105.88 ± 7.59 a	214.13 ± 19.75 b
	Cinnamon soil	CK	8.03 ± 0.10 a	0.21 ± 0.02 a	3.01 ± 0.42 b	0.53 ± 0.06 a	0.76 ± 0.07 a	57.27 ± 5.23 a	13.68 ± 2.04 b	95.86 b ± 8.54 b
		WBC	8.05 ± 0.07 a	0.22 ± 0.05 a	4.57 ± 0.24 a	0.53 ± 0.04 a	0.79 ± 0.10 a	64.63 ± 4.21 b	19.35 ± 3.14 a	162.88 ± 11.22 a
Long-term effect	Fluvo-aquic soil	CK	7.99 ± 0.09 a	0.49 ± 0.08 b	7.41 ± 0.83 b	0.78 ± 0.01 c	1.18 ± 0.12 a	107.39 ± 7.26 b	86.63 ± 5.83 c	108.59 ± 23.35 b
		WBC	8.18 ± 0.10 a	0.85 ± 0.13 a	8.61 ± 0.79 a	0.82 ± 0.03 b	1.35 ± 0.16 a	143.33 ± 18.52 a	129.16 ± 10.27 a	225.82 ± 41.06 a
		MBC	8.30 ± 0.23 a	0.75 ± 0.07 a	8.62 ± 0.89 a	0.85 ± 0.03 a	1.22 ± 0.07 a	133.04 ± 10.47 a	98.02 ± 7.27 b	219.48 ± 26.73 a
	Cinnamon soil	CK	7.96 ± 0.17 a	0.17 ± 0.03 b	2.90 ± 0.37 b	0.54 ± 0.03 a	0.78 ± 0.06 a	46.95 ± 8.06 b	14.95 ± 1.90 b	87.28 ± 15.28 b
		WBC	8.02 ± 0.09 a	0.28 ± 0.04 a	4.02 ± 0.35 a	0.55 ± 0.05 a	0.86 ± 0.07 a	71.48 ± 10.85 a	17.24 ± 2.55 a	146.18 ± 15.52 a

Note: Short-term effect means the results after one crop season; long-term effect means the results sampling in 2019. Different lowercase letters represent the significant difference between treatments ($p < 0.05$).

Table 4. Effect of biochar on the activities of C, N, and P cycling related extracellular enzymes in two calcareous soils.

Soil	Treatment	C-Cycling Soil Enzymes			N-Cycling Soil Enzymes		P-Cycling Soil Enzymes	
		BX (IU L^{-1})	BG (IU L^{-1})	CBH (IU L^{-1})	NAG (U mL^{-1})	LAP (U L^{-1})	ALP (IU L^{-1})	ACP (IU L^{-1})
Fluvo-aquic soil	CK	161.96 ± 17.57 b	45.72 ± 4.16 a	77.30 ± 12.07 b	31.82 ± 6.79 a	134.94 ± 15.83 b	58.97 ± 6.40 b	44.54 ± 4.49 a
	WBC	205.42 ± 16.95 a	46.88 ± 6.59 a	99.36 ± 8.18a	36.82 ± 5.71 a	174.93 ± 3.17 a	73.04 ± 5.20 a	53.13 ± 5.57 a
	MBC	180.86 ± 26.51 ab	52.08 ± 6.32 a	95.56 ± 7.82 ab	37.72 ± 3.38 a	167.16 ± 17.84 a	71.68 ± 4.51 a	50.66 ± 5.09 a
Cinnamon soil	CK	176.58 ± 16.30 b	44.51 ± 11.10 a	84.89 ± 11.41 a	35.76 ± 6.53 a	144.14 ± 23.43 b	59.39 ± 6.14 b	42.16 ± 3.18 a
	WBC	205.61 ± 7.02 a	45.20 ± 6.69 a	104.03 ± 9.49 a	41.02 ± 8.01 a	183.66 ± 9.09 a	73.77 ± 5.82 a	46.09 ± 5.52 a

Note: Different lowercase letters represent the significant difference between treatments ($p < 0.05$).

3.3. Soil Phosphorus Availability

3.3.1. Soil Phosphorus Fraction

Biochar amendment demonstrated a sustained impact on P availability following a single application in calcareous soils (Table 5). In Cinnamon soil, biochar treatment significantly increased P fractions of Ca_2-P, Ca_8-P, and Al-P content and had relatively minimal influence on the stable fractions compared to CK. In Fluvo-aquic soil, WBC significantly increased the Ca_2-P, Al-P, and Fe-P, as well as the stable Ca_{10}-P content. However, it had no significant effect on Ca_8-P and O-P contents. Conversely, MBC treatment had a minimal effect on soil P fractions. These results suggest that WBC amendment was better in enhancing inorganic P availability in calcareous soils.

Table 5. Effect of biochar amendment on soil inorganic P fractions.

Soil	Treatment	Ca_2-P (mg kg^{-1})	Ca_8-P (mg kg^{-1})	Al-P (mg kg^{-1})	Fe-P (mg kg^{-1})	O-P (mg kg^{-1})	Ca_{10}-P (mg kg^{-1})
Fluvo-aquic soil	CK	90.45 ± 8.73 b	116.0 ± 7.0 a	54.19 ± 5.98 b	34.46 ± 8.16 b	156.9 ± 9.5 b	221.3 ± 19.0 b
	WBC	113.14 ± 11.82 a	120.4 ± 12.1 a	67.02 ± 9.12 a	45.36 ± 5.72 a	175.1 ± 8.7 ab	266.6 ± 36.7 a
	MBC	96.83 ± 15.13 b	117.1 ± 4.7 a	62.34 ± 4.04 ab	40.24 ± 7.36 ab	179.3 ± 12.9 a	247.7 ± 22.2 ab
Cinnamon soil	CK	19.86 ± 3.39 b	72.2 ± 11.7 b	29.39 ± 5.03 b	17.56 ± 3.29 a	34.5 ± 1.8 b	153.9 ± 14.4 a
	WBC	30.79 ± 5.23 a	141.9 ± 28.6 a	47.04 ± 3.88 a	20.56 ± 3.97 a	41.2 ± 1.6 a	191.5 ± 22.9 a

Note: Ca_2-P represents monocalcium phosphate [$Ca(H_2PO_4)_2$] and dicalcium phosphate [$CaHPO_4 \cdot nH_2O$]. The Ca_8-P represents phosphates similar to $Ca_8H_2(PO_4)_6 \cdot nH_2O$. The Al-P, Fe-P, and O-P are Al phosphates, Fe phosphates, and Occluded phosphorus fractions. The Ca_{10}-P represents phosphates similar to $Ca_{10}(PO_4)_6 \cdot (OH)_2$. Different lowercase letters represent the significant difference between treatments ($p < 0.05$).

Figure 2 presents the solution ^{31}P-NMR spectra of Fluvo-aquic soil and Cinnamon soil under different treatments. Orthophosphate was the predominant P form in both calcareous soils, making up 88.1–92.9% of the content in Fluvo-aquic soil and 79.7–82.0% in Cinnamon soil (Table 6). Organic P primarily existed as other monoesters and IHP in Fluvo-aquic soil, as well as IHP in Cinnamon soil. In Fluvo-aquic soil, WBC and MBC increased the inorganic P content: WBC increased orthophosphate and polyphosphates by 32.2% and 40.4%, and MBC increased these by 67.2% and 94.5%, respectively, compared to CK. Changes in the organic P fractions varied: in Fluvo-aquic soil, WBC treatment raised the phosphonate content but reduced the IHP and other monoester contents, whereas MBC treatment enhanced the IHP and other monoester contents. In Cinnamon soil, a substantial increase in orthophosphate content was observed, but the organic P fractions remained similar between WBC and CK.

Table 6. Content of P fractions calculated by ^{31}P-NMR of two calcareous soils.

Phosphorus Form (mg kg^{-1})		Fluvo-Aquic Soil			Cinnamon Soil	
		CK	WBC	MBC	CK	WBC
Inorganic phosphorus	Orthophosphate	60.719	80.294	101.540	18.020	21.570
	Polyphosphates	0.453	0.636	0.881	--	--
Organic phosphorus	Phosphonate	0.154	0.304	--	1.954	1.903
	Orthophosphate diesters	0.820	0.604	--	--	--
	Choline phosphate	0.915	--	--	--	--
	IHP	2.578	1.915	2.945	2.365	2.558
	Other monoesters	3.278	2.662	5.338	0.278	0.279

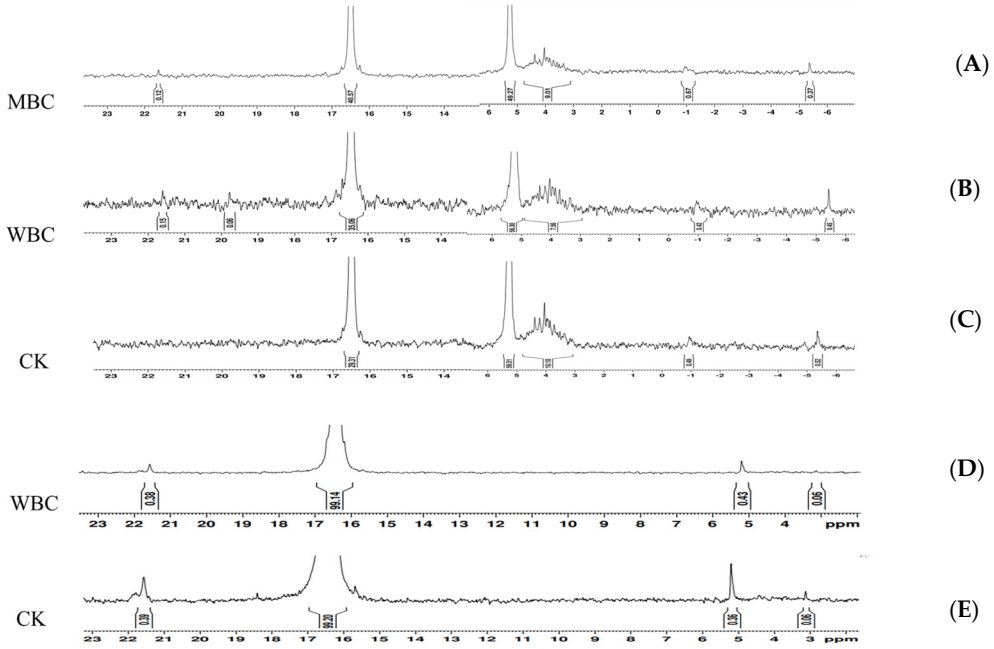


Figure 2. ^{31}P-NMR spectrum of different treatments of two calcareous soils. Note: (**A**–**C**) represent the spectrum of MBC, WBC, and CK in Fluvo-aquic soil, and (**D**,**E**) represent the spectrum of WBC and CK in Cinnamon soil, respectively.

3.3.2. Soil Phosphorus Functional Genes

In Fluvo-aquic soil, both MBC and WBC treatments significantly increased the *phoD* gene copy numbers by 42.0% and 55.4%, respectively, compared to CK (Figure 3). The *phoX* gene copy numbers also increased with MBC and WBC treatments by 33.2% and 20.1%. However, no difference was observed for the *nifH* gene between treatments. In Cinnamon soil, biochar amendment significantly increased the copy numbers of *phoD* and *nifH* genes by 42.3% and 43.3%, but no effect on the *phoX* gene, compared to CK. Thus, biochar exhibited a positive effect on soil P gene copy number in calcareous soils.

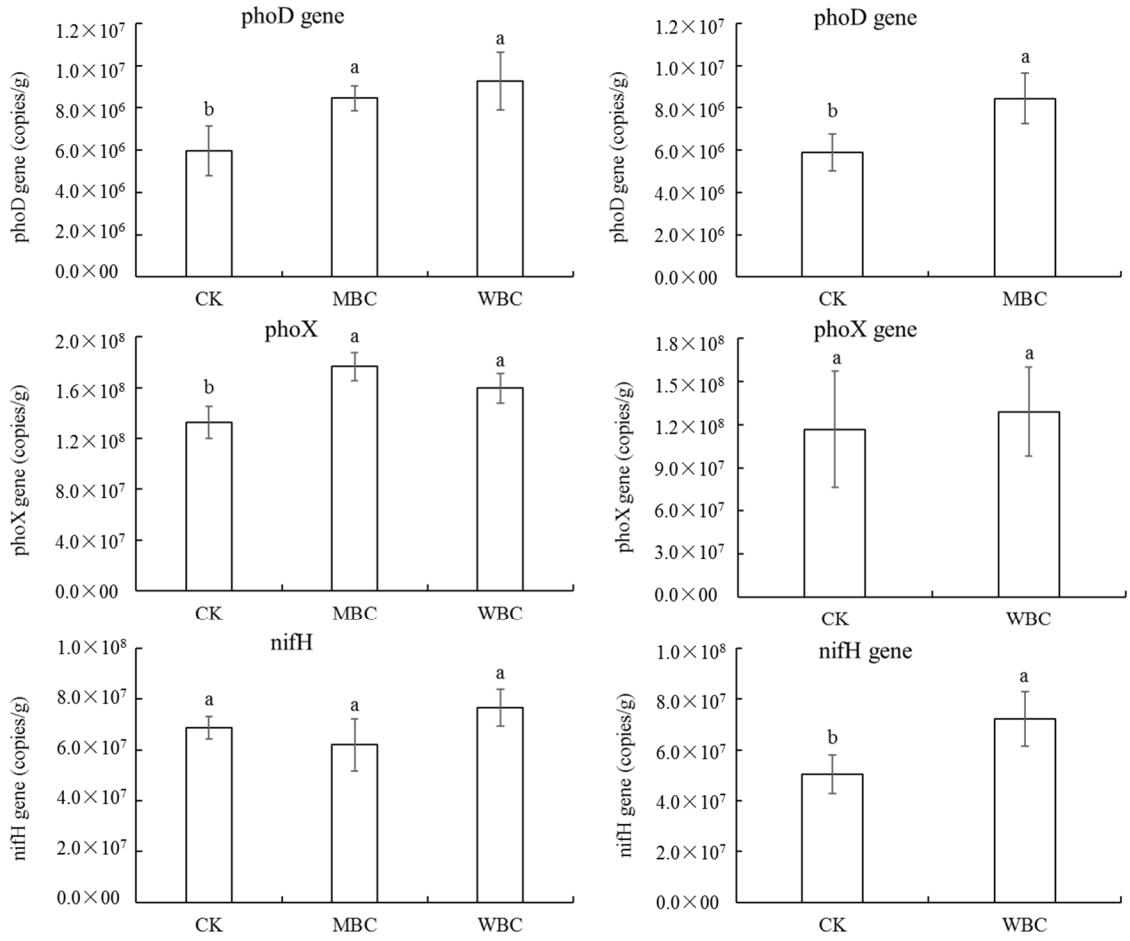


Figure 3. Quantification of P-functional genes in calcareous soils treated by biochar (**left**): Fluvo-aquic soil; (**right**): Cinnamon soil. Note: Different lowercase letters on the column represent the significant difference between treatments ($p < 0.05$).

3.4. SEM Imaging and EDS Spectra of Fresh and Aged Biochar

The regulation effect of biochar on soil fertility was related to the aging process of biochar in soil. Both WBC and MBC still retained some of the original morphological features after several years of aging (Figure 4). The surfaces of the fresh WBC and MBC appear relatively smooth, featuring pores or holes in the biochar particles. The pore structure of fresh WBC was comparatively regular, exhibiting greater porosity than fresh MBC, whose pore size distribution was less uniform. After aging, the pore structure of the biochar surface was somewhat damaged, with some pores or holes blocked. The surface appeared rough, with several small particles adhered. The EDS results revealed that the surface element composition of fresh biochar mainly consisted of C, O, and Si. Following several years of aging, the elemental composition tended to be complex, especially the enrichment of mineral elements of P, K, Ca, Mg, and others. These results suggest that biochar can still regulate soil mineral nutrients during the aging process.

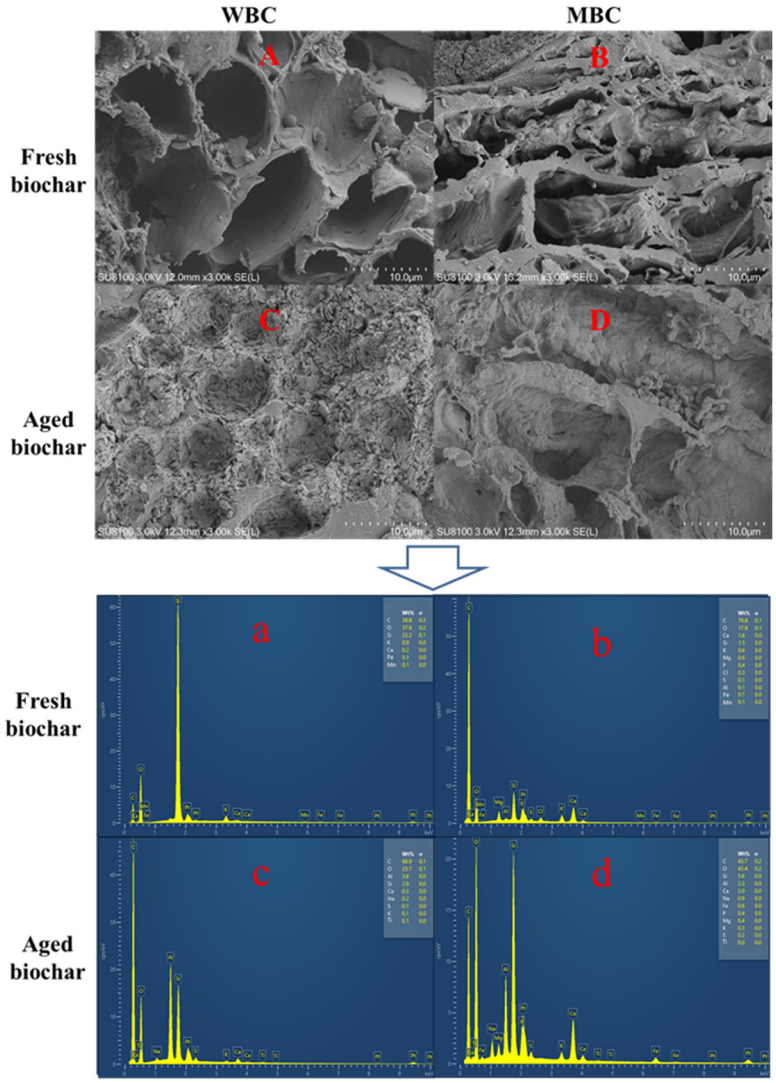


Figure 4. Scanning electron microscope (SEM) images and the EDS map of mineral elements for fresh and aged biochar. Note: (**A**,**B**) represent the SEM images of fresh WBC and MBC; (**C**,**D**) represent the aged WBC and MBC; (**a**,**b**) represent the EDS maps of fresh WBC and MBC, (**c**,**d**) represent the aged WBC and MBC, respectively.

4. Discussion

4.1. Long-Term Effect of Biochar on the Productivity of Calcareous Soils

The effect of biochar application on crop productivity enhancement, particularly the long-term and variable effects, has been extensively documented [25,36]. This study also revealed a sustained increase in crop yield with biochar application in two calcareous soils. This finding aligns with numerous other long-term studies. For instance, Jones et al. [37] reported from a 3-year experiment that biochar significantly boosts grass crops above-ground biomass. Kätterer et al. [38] found that biochar application consistently increased the yields of maize-soybean rotations in a 10-year field experiment across three sites in Kenya. Conversely, other studies have reported that biochar has no long-term effect on crop yield. For example, Haider et al. [39] conducted a field experiment in a temperate sandy soil and found no significant effect on crop yield over four years. Liu et al. [40] found that biochar improved soil fertility for at least six years, but it did not result in an increased crop yield in rice paddies. These variations may be related to the soil conditions and biochar properties. In this study, the yield-enhancing effect of biochar was more pronounced for WBC than MBC and more beneficial in Fluvo-aquic soil than Cinnamon soil. Therefore, the

long-term benefits of biochar on crop productivity varied, but they were demonstrated in the calcareous soils in the current study.

The long-lasting impact of biochar addition on crop productivity can be primarily attributed to the sustained improvement in soil fertility. This study showed that biochar significantly improved soil fertility five or six years after a single application in two calcareous soils, including an increase in SOC and available nutrient content and promotion of soil aggregate formation. This aligns with Kätterer et al. [38] that biochar soil amendments persistently enhanced soil fertility over ten years, based on field experiments across different sites within the maize-soybean rotation system, leading to an increase in productivity. The initial addition of biochar introduces a substantial amount of organic matter and ions into the soil [41], which plays a pivotal role in soil fertility in the short term. For example, the available P and K content increased more in the short-term results of this study. The long-term mechanism may be related to the stable structure of biochar and the interplay between microbial abundance and soil aggregation. The microbe excretions promote soil aggregation [42]. This mechanism was highlighted by our study, in which biochar retained its base structure for several years, with an enrichment of mineral elements attached to its surface. Consequently, biochar significantly increased the MWD of soil aggregates by 53.1–73.5% in Fluvo-aquic soil and 64.7% in Cinnamon soil. In turn, the improved soil structure is beneficial to microorganisms. Studies by Burrell et al. [43] suggested that biochar amendments have a long-term effect on soil aggregate stability, permanently improving soil water availability and porosity.

Biochar amendment creates a better soil structure and nutrient conditions for microbial activity, thereby facilitating benefits such as nutrient acquisition and crop productivity. In our study, biochar soil amendment significantly increased the extracellular enzyme activity related to C, N, and P cycling in two types of calcareous soils. The increases in extracellular enzyme activity with biochar treatment have been widely reported. Pokharel et al. [9] summarized the related studies and indicated that biochar enhances the activity of enzymes related to N and P cycling, especially under calcareous soil conditions. In comparison, wide variations exist between soil conditions. A six-year field experiment conducted in acidic, loamy sand soil showed that biochar amendment increases in soil enzymatic activity related to nutrient cycling in successive years [44]. However, Ameloot et al. [45] found that biochar treatment showed a lower or equal soil microbial activity several years after application in neutral soil. Our study highlights a positive long-term effect on soil enzyme activity in two types of calcareous soils. Therefore, the long-term increase in soil enzyme activity related to nutrient cycling due to biochar application presents a promising strategy for ensuring the sustainability of agricultural production.

4.2. Long-Term Effect of Biochar on Phosphorus Availability in Calcareous Soils

Our study revealed a long-term increase in soil P availability with biochar treatment in two calcareous soils. The short-term increase effect is mainly related to the direct input of P from biochar. However, the long-term effect is not solely due to the P supplied but is also related to the release of stable soil P resulting from interactions between biochar and soil [46], which lead to the transformation of P forms [47]. In this study, we used continuous extraction and ^{31}P-NMR spectroscopy to measure the P fractions in the two calcareous soils and found that biochar increased the proportion of labile P fractions such as Ca_2-P, Al-P, orthophosphate, and polyphosphates, which indicating increased mobility of stable P forms in calcareous soil under long-term conditions. In addition to this, biological or abiotic factors are involved in regulating P fractions, especially in the long term.

The effect of biochar on the transformation of soil P forms is initially attributed to changes in soil pH. Although biochar did not significantly affect the bulk soil pH of the

two calcareous soils in this study, the pH difference between biochar and soil produced a strong pH gradient at the biochar–soil interface, which affected P availability [48]. This interface pH difference reduces the concentrations of free cations, such as Ca^{2+}, $Al^{3+,}$ and Fe^{3+}, thereby reducing the precipitation of mineral phosphates. Hence, the long-term effect of biochar treatment on soil P availability may be due to physicochemical regulation processes. Biochar can adsorb organic molecules on its surface, chelate cations, and attract polar or non-polar organic molecules, which reduce the transformation of soluble P into stable macromolecular complex P forms [49]. As biochar ages in the field, the hydrophilic oxygen-containing functional groups on the biochar's surface increase [5]. The SEM results of this study found that biochar still showed the ability to adsorb ions after several years in calcareous soils. Moreover, previous studies indicated that long-term P recycling might be influenced by the CEC of the biochar [41], which would raise the free cation content and maintain soil P nutrients, which could be one of the persistent sources of soil available P.

The long-term effect of biochar on P transformation in calcareous soil also results from biological mechanisms [41,50]. Biochar amendment alters soil microbial activity [51]. Studies have shown that biochar amendment could increase soil phosphatase activity [9,52], causing the release of organic compounds or mineral complex P fractions in the soil. The current study also found that biochar amendment significantly increased soil alkaline phosphatase activity in both Fluvo-aquic and Cinnamon soils. The activity of soil-alkaline phosphomonoesterase positively correlates with the copy number of P-functional genes, such as the *phoD* gene [53]. We also measured these gene copy numbers and found that biochar soil amendment increased the soil P gene copy numbers in calcareous soils for the *phoD*, *phoX*, and *nifH* genes, all of which are known to encode alkaline phosphomonoesterase. This result aligns with the finding of Tian et al. [54] that biochar fosters a more organized *phoD* gene community, enhances alkaline phosphatase activity, and promotes organic P mineralization under low P input.

5. Conclusions

Biochar amendment was shown to cause a long-term benefit on crop productivity in calcareous soil. A persistent crop yield increase was observed with biochar amendment, although this varied with different soil conditions and biochar types. Soil fertility was consistently improved under biochar treatment, with significant increases in the SOC soil's available nutrient content. The carbon, nitrogen, and phosphorus cycling-related extracellular enzyme activity was increased for five or six years following one-time biochar application. This study also demonstrated a long-term increase in soil phosphorus availability in both Fluvo-aquic soil and Cinnamon soil. The increased available phosphorus was from the transformation of soil phosphorus fraction, as the relatively labile phosphorus fractions were elevated, such as Ca_2-P, Al-P, orthophosphate, and polyphosphates. This long-term increase in soil phosphorus availability may be related to biological processes, as biochar amendment significantly increased the alkaline phosphatase activity and raised the *phoD*, *phoX*, and *nifH* gene copy numbers. This study confirmed the long-term benefit of biochar on nutrient regulation in calcareous soils, but the effects in other soil conditions need further study.

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